INCH-POUND

MIL-T-27/102A
13 March 1991
SUPERSEDING
MIL-T-27/102
2 AUGUST 1976

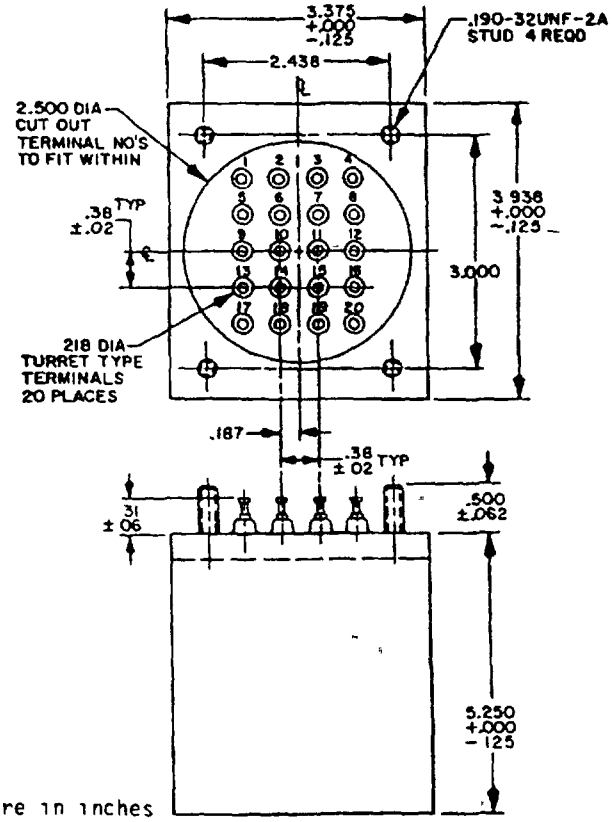
MILITARY SPECIFICATION SHEET

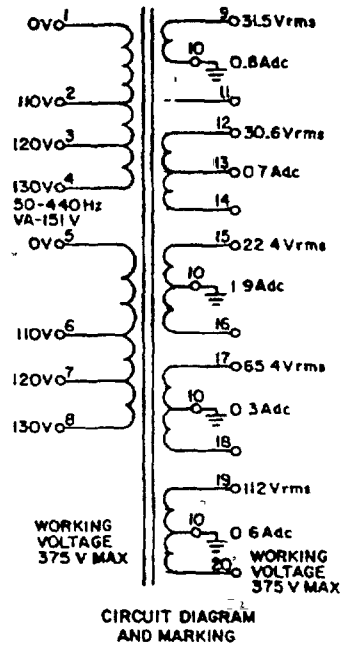
TRANSFORMER, POWER, STEPDOWN, TF4R03KA921

Ⓐ Inactive for new design after the date of this specification.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-27.





CIRCUIT DIAGRAM AND MARKING

Inches	mm
.02	0.51
.06	1.52
.062	1 57
.125	3.18
.187	4.75
.190	4.83
.218	5.54
.31	7 9
.38	9.7
.500	12.70
2.438	61.92
2.500	63.50
3 000	76.20
3.375	85.72
3.938	100.02
5.250	133.35

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± 010 (0.25 mm).
4. Marking shall be on the side of the case.

FIGURE 1 Dimensions and configurations.

Ⓐ denotes changes

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

Primary voltage. (1-3) (5-7) 120 volts rms ±10, 50 Hz to 440 Hz.

Secondary voltage:

(9-11) 31.5 volts rms, 0.80 A dc.
(12-14). 30.6 volts rms, 0.7 A dc.
(15-16): 22.4 volts rms, 1.90 A dc.
(17-18). 65 4 volts rms, 0.3 A dc.
(19-20) 112 volts rms, 0.60 A dc.

DC resistance

(1-4) 4.5 ohms maximum.
(5-8): 5.0 ohms maximum.
(5-6)(9-11): 0.47 ohm maximum.
(12-14): 0 68 ohm maximum.
(15-16). 0 14 ohm maximum.
(17-18) 2.07 ohms maximum.
(19-20): 3.62 ohms maximum.

Working voltage:

(1-8): 375 volts maximum.
(9-20): 375 volts maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal.

Material Steel.

Terminals: Solder, turret type.

Terminal height. 0.310 inch ±0.600.

Weight 10 0 pounds.

Operating temperature range -55°C to +85°C.

Terminal strength MIL-STD-202, method 211, test condition A, 5.0 pounds.

Dielectric withstanding voltage

At sea level 1050 volts rms.

At barometric pressure 945 volts rms.

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Electrical characteristics:

No load: With 120 volts rms, 60 Hz across (1-3) (5-7) in parallel, current in (1-3) shall not exceed 175 milliamperes and the power in (1-3) shall not exceed 4.8 watts.

Voltage across (9-11): 33.0 volts rms ct, ±5 percent.
Voltage across (12-14): 32.0 volts rms ct, ±5 percent
Voltage across (15-16). 24.0 volts rms ct, ±5 percent.
Voltage across (17-18): 70.0 volts rms ct, ±5 percent.
Voltage across (19-20): 115 volts rms ct, ±5 percent.

Rated load: With 120 volts rms, 60 Hz across (1-3) (5-7) in parallel.

Voltage across (9-11). 31.5 volts rms, ±5 percent.
Voltage across (12-14): 30.6 volts rms, ±5 percent.
Voltage across (15-16): 22.4 volts rms, ±5 percent.
Voltage across (17-18). 65.4 volts rms.
Voltage across (19-20): 112 volts rms.

Temperature rise: +28.73°C with 120 volts rms, 60 Hz across (1-3) at an ambient temperature of +65°C.

Ⓐ Quality assurance provisions:

Qualification inspection: Not applicable.

Quality conformance inspection: Groups A and B tests of MIL-T-27 shall be applicable.

Marking location: See figure 1.

Part number: M27/102-01.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS, SH
Air Force - 17, 99
DLA - ES

User activities.
Army - AR, ME
Navy - AS, MC
Air Force - 19

Preparing activity.
Army - ER

Agent:
DLA - ES

(Project 5950-0753-32)

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